Computational geometry exercises

A10-P01 Orientation and sign translation

Let a=(1,1), b=(5,2), c=(3,4), q=(9,3). Evaluate orient(a,b,c), orient(a,c,b), and orient(a,b,q) term by term, and give triangle abc's area. How do signs and the left-turn condition change for CLRS DIRECTION? What visual reversal occurs if a downward screen y-axis is used without coordinate conversion?

A10-P02 Boundary cases are part of the specification

Fix ab:(0,0)→(6,0). For every cd below give o1,o2,o3,o4, the closed-intersection answer, the proper-crossing answer, and a successful on test if any.

Case	c	d
I	(2,0)	(8,0)
II	(7,−1)	(7,2)
III	(6,0)	(6,0)
IV	(3,−2)	(3,2)

Also test distinct point segments (1,1)→(1,1) and (1,2)→(1,2). Why is observing four zero determinants insufficient?

A10-P03 Arithmetic and comparators

Derive the conservative orientation bound $8B^2$ for coordinates of absolute value at most B. With B=100000, is signed 32-bit arithmetic sufficient? Why must widening precede subtraction? Why does a fixed zero epsilon lack a scale-independent guarantee? Use 0, 0.75ε, and 1.5ε to show that “distance less than ε means equal” is not transitive.

A10-P04 Deletion exposes an intersection

Use a:(0,0)→(10,5), c:(1,2)→(3,2), b:(2,6)→(9,0). List bottom-to-top status and each newly tested pair at every processed endpoint. Compute the exact a,b intersection. What happens if tests occur only at insertion? Why can detection return before reaching the crossing's x-coordinate?

A10-P05 Tied events, verticals, and shared points

How should x=2 events be ordered for a:(0,1)→(2,1), b:(2,1)→(4,2)? In a separate status with heights 1,2,6 at x=2, which heights do vertical intervals [3,4] and [2,4] hit? Explain insertion, query, and deletion order at one x, plus checks among vertical segments there. Does adding an id tie-break alone prove correctness for three segments through one point?

A10-P06 Detection is not reporting

a:(0,0)→(4,4), b:(0,4)→(4,0), c:(0,5)→(4,5). Compare status at x=1 and x=3. Which invariant fails if `return true` is replaced by printing and continuing? What additional events and updates does reporting require? Construct $\Theta(n^2)$ intersections using m horizontal and m vertical segments; give n and k.

A10-H01 Polar-angle ties

List the raw polar order and retained points for the hull input. Prove that keeping the farthest point on each ray preserves the hull.

A10-H02 Replay the stack

Write every stack update and the cross products tested at D and F.

A10-H03 Collinear output

What does this chapter return for (0,0),(1,0),(2,0) with a repeated (1,0)? Can changing only the while test return all boundary points?

A10-H04 Choose a hull method

Compare Graham and Jarvis for n=10^6 with h=4 and h=n.

A10-H05 Split repeated x values

Give X, the side ids, and both Y arrays for the closest-pair input. Explain why x≤median is unsafe.

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A10-H06 Compute the strip

Compute the two recursive squared distances, δ, strip order, and candidates passing the original vertical threshold.

A10-H07 Prove the successor bound

Derive seven successors using a 2δ by δ rectangle. State assumptions for same-side distances, line points, and δ=0.

A10-H08 Repair the complexity and duplicates

What happens if every call sorts by y? Can closest pair deduplicate equal coordinates first?

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd ed., Chapter 33. MIT 6.046J Fall 2004 computational geometry lecture slides, segment intersection and closest pair. Original handwritten algorithms notes (演算法_wjungle), computational geometry pages.

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